



Bilateral Fisheries Agreements

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Introduction

- ▶ Pacific fisheries licenses are levied on effort (per vessel or vessel day)
- ▶ Access fees are determined through either bilateral or multi-lateral negotiation (US tuna treaty)
- ▶ Coastal states are interested in developing their own industry, revenue through licenses and access fees and employment and onshore development (processing - loining or canning).
- ▶ Other revenue sources include transshipment fees and miscellaneous admin fees, observer fees (cost-recovery).



Literature Review

- ▶ Previous economics literature focuses on multilateral agreements
- ▶ Legal literature has investigated bilateral agreements
- ▶ Economic literature on access pricing in regulatory economics and industrial economics has had little influence on fisheries economics
- ▶ Literature on coastal and distant water fisheries nation relations doesn't get to the heart of the issues



Literature

- ▶ Munro, G. R. (1982). Cooperative fisheries arrangements between Pacific coastal states and distant water nations. In: H.E. English and A. Scott (Eds.) Renewable Resources in the Pacific Proceedings of the 12th Pacific Trade and Development Conference, held in Vancouver, Canada, 7-11 September 1981



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Example: Cook Islands

Part B: Fees for Fishing within Fishery Waters

These fees apply where there are no fees established by an applicable bilateral or multilateral access agreement.

5. Annual fishing licence for Cook Islands Fishing Vessels –

	Locally-Based	Not Locally-Based
Under 20 metres LOA	\$2,000	
20 – 40 metres LOA	\$5,000	\$40,000*
Over 40 metres LOA	\$7,500	\$60,000*

The requirements for a vessel to be considered “locally-based” are –

- (a) the vessel is fully operated from a Cook Islands port;
- (b) the vessel fishes mainly in Cook Islands waters; and
- (c) the vessel lands all or the majority of its catch in Cook Islands.

6. Annual fishing licence for Foreign Fishing Vessels

- (a) Under 40 metres LOA \$50,000*
- (b) Over 40 metres LOA \$75,000*

7. Annual fishery development fee \$10,000

This additional fee is payable by all vessels not locally-based

8. Fees in Part B marked with an (*) should be reduced by \$20,000 if the following conditions are met:

- (a) in the licence period immediately preceding the licence period which is the subject of the present application, the vessel unloaded/landed twenty metric tonnes or more of fish products caught in Cook Islands fishery waters in a Cook Islands port; and
- (b) that product was processed through a local fish processing facility approved by the Secretary of Marine Resources.



Model

- ▶ Starting point is the standard Gordon-Schaefer model
- ▶ We consider a game between domestic fleets and foreign fleets
- ▶ the game is solved parametrically based on a fixed license fee, i.e. the Nash equilibrium of the game depends on the license fees levied
- ▶ A social planner (the national fisheries regulator) maximizes their objective, a social welfare function by choosing the license fee to set and number of licenses to issue. To set both correctly knowledge of the demand function for licenses is required.
- ▶ Once license fees have been set a two-part pricing approach can be used to set an access fee.
- ▶ Two-part pricing would extract all the rent from foreign vessels, so they might simply choose not to fish within the EEZ.
- ▶ This leads to a bilateral bargaining between coastal and distant water states/fleets.



Set-up

$$\frac{dX(t)}{dt} = rX\left(1 - \frac{X}{K}\right) - qXe_D - qXe_F \quad (1)$$

- ▶ where X is the stock of fish r is the intrinsic growth rate of the stock,
- ▶ K the carrying capacity or the steady-state level of unfished biomass,
- ▶ q catchability which is assumed to be homogeneous for both fleets due to them fishing the same stock with the same gear.
- ▶ e_D is the fishing effort of the domestic fleet and
- ▶ e_F the fishing effort of the foreign fleet.

Steady-state

We will assume a steady-state in what follows so that

$$\frac{dX(t)}{dt} = rX\left(1 - \frac{X}{K}\right) - qXe_D - qXe_F = 0 \quad (2)$$

Solving for X we obtain the steady-state level of the stock

$$X^*(e_F, e_D) = \begin{cases} 0, \\ \frac{K}{r}(r - qe_D^* - qe_F^*) \end{cases} \quad (3)$$

Fleet Rents - Strategic game

The two fleets compete for a common resource. The regulator can partially influence the cost of fishing by setting license fees.

$$\max_{e_F} \Pi_F = pqX^*(e_F, e_D)e_F - c_F e_F - f_F e_F \quad (4)$$

$$\max_{e_D} \Pi_D = pqX^*(e_F, e_D)e_D - c_D e_D - f_D e_D \quad (5)$$



Nash equilibrium

In equilibrium the foreign fleet exerts the following level of fishing effort:

$$e_F = \frac{(Kpqr + c_D r - 2c_F r + f_D r - 2f_F r)}{(3Kpq^2)} \quad (6)$$

and the domestic fleet exerts the following level of fishing effort:

$$e_D = \frac{(Kpqr - 2c_D r + c_F r - 2f_D r + f_F r)}{(3Kpq^2)} \quad (7)$$

Welfare maximization

We use the following social welfare function that combines the rent of the domestic fleet and the license fee revenue. more weight is given to domestic license fee revenue than foreign license fee revenue to avoid domestic revenue being welfare neutral.

$$W = \Pi_D + 2f_D e_D + f_F e_F \quad (8)$$

Recall that $P_D = pqXe_D - c_D e_D - f_D e_D$, so that

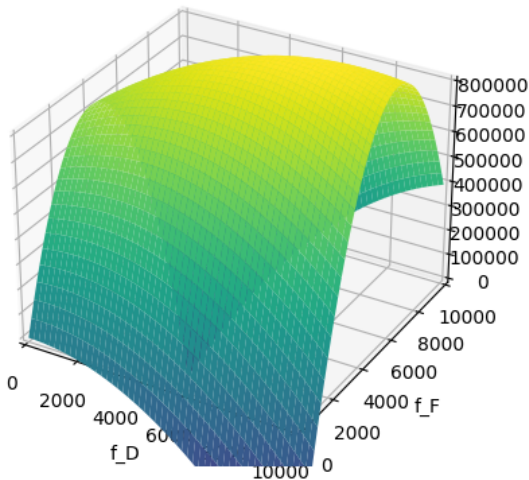
$$W = pqXe_D - c_D e_D - f_D e_D + 2f_D e_D + f_F e_F \quad (9)$$

$$= pqXe_D - c_D e_D + f_D e_D + f_F e_F \quad (10)$$



Social Welfare function

Surface plot





Solution

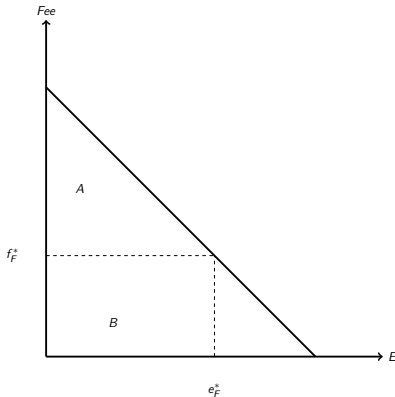
$$f_D^* = Kpq/3 - c_D/3$$

$$f_F^* = 2Kpq/3 - 4c_D/15 - 2c_F/5$$



Two-part Tariff

Two-part pricing and access fees (assuming a fixed domestic fee):



Triangle A is fixed component of the two-part tariff corresponding to access fees. Rectangle B is the license fee revenue at the optimal fee f_F^* . The Diagram is adapted from Lewis (1941).



Access Fee (without concessions)

$$\text{Access Fee} = \frac{1}{2}(\text{Fee at zero demand} - \text{Optimal Foreign Fee}) \\ \times \text{Foreign License Demand at Optimal Foreign Fee}$$

This is the triangle A.

Bilateral Bargaining

Two standard solutions are the Nash bargaining solution and Kalai-Smorodinsky.

The Nash solution is found by maximising the Nash product

$$\max_{(\theta)} (U_D(E, f) - d_D)(U_F(E, f) - d_F) \quad (11)$$

where

$U_F(E, f) = \frac{1}{2}(f_F(e_F)|_{e_F=0} - e_F^*)e_F^*$ or the area A . The coastal state wishes to determine to obtain some portion θ of this and the distant-water fleet the portion $1 - \theta$. So,
 $U_D(E, f) = \theta \frac{1}{2}(f_F(e_F)|_{e_F=0} - e_F^*)e_F^*$ and
 $U_F(E, f) = (1 - \theta) \frac{1}{2}(f_F(e_F)|_{e_F=0} - e_F^*)e_F^*$



Nash bargaining solution

The Nash solution is found by maximising the Nash product

$$\begin{aligned} & \max_{\theta} (\theta A - f_D^* e_D^*) (1 - \theta) A \\ & = \theta(1 - \theta) A^2 - f_D^* e_D^* (1 - \theta) A \end{aligned}$$

where $A = \frac{1}{2}(f_F(e_F)|_{e_F=0} - e_F^*)e_F^*$ and assuming the disagreement point $(d_D, d_F) = (f_D^* e_D^*, 0)$. This needs defending!



Disagreement points

- ▶ The coastal states outside option in the event of the failure to agree is simply the domestic license fee revenue.
- ▶ The distant water states disagreement point depends on its outside options
- ▶ The fleet could fish in another zone or it could fish on the high seas.
- ▶ To introduce other zones requires a multilateral model (work in progress)
- ▶ The most plausible outside option is to fish on the high seas which under free-entry implies full rent dissipation and the disagreement payoff is zero. I don't model the high seas explicitly but one can motivate in zone fishing by arguing that the cost of fishing on the high seas is higher or catchability is lower.



Nash Bargaining Solution

The first-order conditions are

$$A^2 - 2\theta A^2 + f_D^* e_D^* = 0$$

or

$$\theta = \frac{A^2 + f_D^* e_D^*}{2A^2} = \frac{1}{2} + \frac{f_D^* e_D^*}{2A^2}$$

where $A = \frac{1}{2}(f_F(e_F)|_{e_F=0} - e_F^*)e_F^*$



Points for Discussion

- ▶ This solution will be the same as the Kalai-Smorodinsky solution as long as the triangle A is fixed in size and fees and the number of vessels being given access are not subject to negotiation. If these are subject to negotiation the two solutions diverge. In many cases the fees are fixed by law and effectively non-negotiable. The number of vessels may be negotiated (so I still need to explore this case).
- ▶ Once a fee is set the demand for licenses uniquely determines license demand. So negotiating license numbers doesn't make much sense unless the demand function is uncertain. In which case a fixed fee doesn't uniquely determine the demand for vessel licenses and these could be negotiated.



Conclusion

- ▶ There is a confusion in the literature between market rent and producer rent. Note these are equivalent in the Gordon-Schaefer model but not the model presented here
- ▶ Coastal state should receive more than half of the rent in the foreign license fee market. $\theta = \frac{1}{2} + \frac{f_D^* e_D^*}{2A^2}$
- ▶ Distant-water states should receive less than half $1 - \theta = \frac{1}{2} - \frac{f_D^* e_D^*}{2A^2}$
- ▶ In other work I've extended the analysis to a multi-zone setting (presented in Copenhagen SURF conference 2023)
- ▶ I've also considered a two-part pricing approach to license and access fees within the pure Gordon-Schaefer framework (presented at CEA Vancouver 2026 and IIFET, Torshavn, 2026)
- ▶ Also working on a review of license laws and regulations in the Pacific (presented in Suva, Fiji, April, 2025).



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Thanks for listening!

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